

• ISM:

(i) Neutral Hydrogen: (H^0 or HI)

- 21 cm radiation, $E = \frac{hc}{\lambda}$
- Caused by a spin-flip $\uparrow \rightleftharpoons \downarrow$
→ QM effect
- 60% of total mass
 $T \approx 80 K$ $\langle E \rangle \approx \frac{f}{2} kT$
 $n_H \approx 3 \times 10^6 m^{-3}$

(ii) Molecular Hydrogen: (H_2) H_2

- dense HI regions $T \approx 30 K$
- $n_{H_2} \approx 10^9 - 10^{12} m^{-3}$
- $\approx 20\%$

*Aside: Note from stat mech on, av. for a simple (non-rel.) system we have

$$E = \frac{f}{2} k_B T$$
$$V_{rms} = \sqrt{\langle v^2 \rangle} = \sqrt{\frac{3k_B T}{m}}$$

from

$$\frac{1}{2} m V_{rms}^2 = \frac{3}{2} k_B T \Rightarrow V_{rms} = \sqrt{\frac{3k_B T}{m_{H_2}}}$$

- Since, $m_{H_2} > m_H$ then we know that on average the gas has particles moving at a slower rate and hence it will have a lower temperature

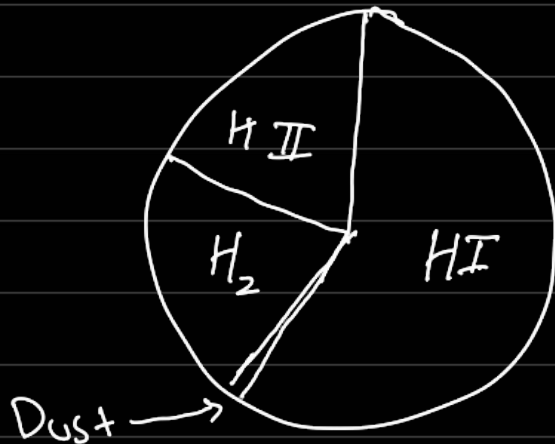
(iii) Ionised Hydrogen: (H^+ or HII)

- Near hot, blue stars (O & B)
- $n \approx 10^8 - 10^{10} m^{-3}$ Near O stars

$$n_e \approx 10^6 \text{ m}^{-3}$$

near B stars &
planetary nebulae

- $T \approx 8000 \text{ K}$
- 20%



- Very energetic: $(V_{rms})_{HII} \gg (V_{rms})_{H2/HI}$

(iv) Dust: Carbonates/Silicates

- $R_{dust} < 1 \mu\text{m}$
- Extinction/emission
- < 1%

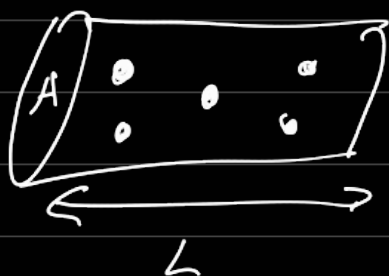
* ISM: 70% H and 28% He
3:1 ratio

Extinction:

- Dust clouds in the ISM

$\sigma \equiv$ cross-section

$$\sigma = \pi r^2$$



$$n = \frac{(\#)}{V} = \frac{N}{V}$$

$$V = AL$$

$$\therefore N = nAL$$

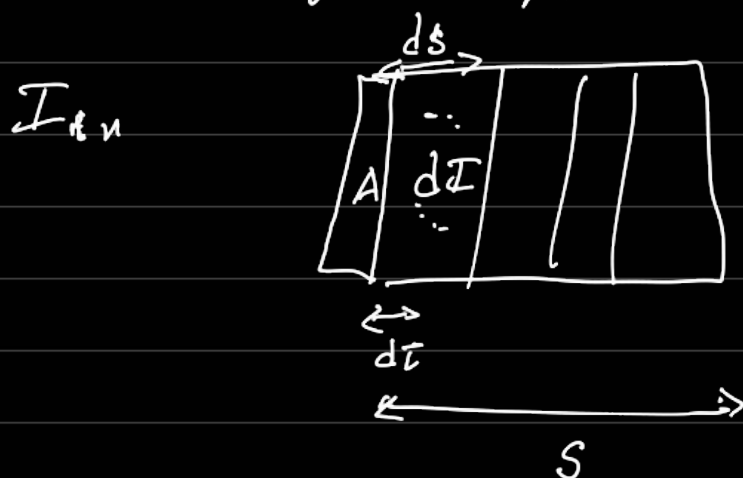
$$\sigma_{tot} = N\sigma = nAL\sigma$$

$$(\frac{\sigma_{tot}}{A} = \frac{\sigma_{tot}}{A} = nL\sigma$$

$$\boxed{\tau \equiv nL\sigma} \quad (\text{optical depth})$$

or frac of light blocked

$\tau \ll 1$: low density, light passes easily
 $\tau \approx 1$: high density, absorbed



$$n = N/V$$

$$N = nV = nA ds$$

$$\rho = mn$$

$$N = \left(\frac{\rho}{m}\right) A ds$$

σ : units of Area

$$\begin{aligned} * \sigma_{tot} &= N\sigma = \left(\frac{\rho}{m}\right) A \sigma ds \\ &= \rho A \left(\frac{\sigma}{m}\right) ds \end{aligned}$$

$$\left(\frac{\sigma}{m}\right) \equiv K_\lambda \quad (\text{absorption coefficient})$$

$$\frac{\sigma_{tot}}{A} = \rho K_\lambda ds \equiv (\text{fraction absorbed})$$

$$* I_{in} \quad ds:$$

$$dI = -I \cdot (\rho K_\lambda ds)$$

minus since the rate of change is negative

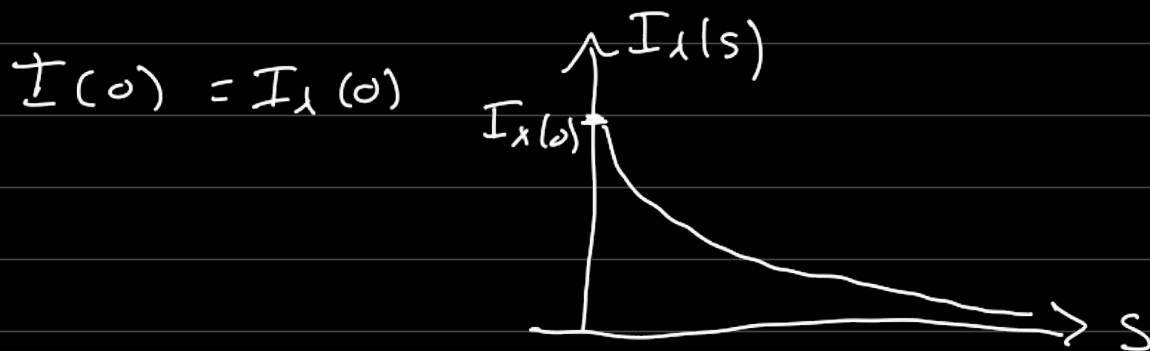
$$dI = -I \rho k_\lambda ds$$

$$= -I d\tau_\lambda$$

(assuming) $d\rho = 0$ (i.e. const. density)

$$I_\lambda(s) = I_\lambda(0) e^{-\rho k_\lambda s} = I_\lambda(0) e^{-\tau_\lambda(s)}$$

• Note: assumes $k_\lambda = \frac{\sigma}{m}$ is the same throughout the layer



* Interstellar absorption was from stat. study of open clusters: lie in the disk of the galaxy
 → stars further in clusters were dimmer



• Dust: responsible for absorption of light

* Must modify: (178)

$$m - M = 5 \log_{10} \left(\frac{d}{10} \right)$$

$$\uparrow \quad \quad \uparrow \quad = 5 \log_{10} |d| - 5$$

rel.

abs.

mag

mag

* Assumes no absorption, so must modify

$$m = M + 5 \log_{10}(d) - 5 + A_\lambda$$

$$m_{\text{obs}} = m_{\text{true}} + A_\lambda$$

• Stars within galactic plane: $A_V \approx 1.5d$
 $\uparrow$ kpc

• V-band:

• absorb more light at shorter wavelengths
 distant stars will appear redder

* Redness of a star: $(B-V)$
 $\rightarrow (B-V)$ continues to increase

$$E(B-V) \equiv \Delta(B-V)$$

$$E(B-V) \approx 1.5d$$

$$I \approx \frac{\mathcal{L}}{4\pi d^2}$$

$$\mathcal{L} = \sigma A T^4$$

$$m_1 - m_2 = 2.5 \log_{10} \left(\frac{I_2}{I_1} \right)$$

$$m_{\text{obs}} - m_{\text{true}} = 2.5 \log_{10} \left(\frac{I_\lambda(0)}{I_\lambda(s)} \right)$$

$\uparrow$ $I_\lambda(0)$

$$I_\lambda(s)$$

$$A_\lambda = -2.5 \log_{10} (e^{-\rho K_\lambda s}) = 1.09 \rho K_\lambda s$$

$$= 1.09 \tau_\lambda$$

* interstellar reddening

$$E(\lambda-V) = A_\lambda - A_V$$

• optical $A_\lambda \propto \frac{1}{\lambda}$

• small dust:

$$A_\lambda \propto \frac{1}{\lambda^4} \leftarrow \text{Earth's}$$

- photometric system:

$$B: 445 \text{ nm}, \quad \Delta\lambda \approx 94 \text{ nm} \quad (\text{Blue})$$

$$V: 551 \text{ nm}, \quad \Delta\lambda \approx 88 \text{ nm} \quad (\text{Visual})$$

- * Interstellar Extinction:

$$E(B-V) = A_B - A_V = A_V \left(\frac{A_B}{A_V} - 1 \right)$$

- In the optical:

$$A_\lambda \approx \frac{1}{\lambda}$$

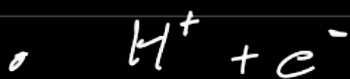
$$E(B-V) = A_V \left(\frac{\lambda_V}{\lambda_B} - 1 \right) = 0.28 A_V$$

$$A_V \approx 3.5 E(B-V)$$

$$R \equiv \frac{A_V}{E(B-V)} \approx 3.2 \pm 0.2$$

$$E(U-B)/E(B-V) = 0.74$$

HII Regions: ionized Hydrogen H^+ : prot +



- QM Hydrogen atom:

$$\left(E_n = -13.6 \text{ eV} \left(\frac{1}{n^2} \right) \right) \quad \begin{matrix} \text{(Binding)} \\ \text{(Energy)} \end{matrix}$$

$$\Delta E = |E_f - E_i| = 13.6 \left[\frac{1}{n_i^2} - \frac{1}{n_f^2} \right]$$

$$n_i = 1, \quad n_f = \infty$$

$$\Delta E = 13.6 \text{ eV}$$

• Ionisation potential: $I_H = 13.6 \text{ eV}$

• Star w/ $T_* \approx 30,000 \text{ K}$

avg. energy

$$\langle E_\gamma \rangle \approx \frac{f}{2} k_B T = k_B T_* = 4 \times 10^{-19} \text{ J} = 0.2 I_H$$

$$[S_*] = \frac{(\# \gamma's)}{\text{sec}} \approx \frac{N}{\Delta t}$$

(i) $N_\gamma = S_* t$

(ii) $N_\gamma = V \cdot n_H$

look at a uniform sphere

$$V = \frac{4}{3} \pi R^3$$

$$N_H = N_\gamma = \frac{4}{3} \pi R^3 n_H$$

thus

$$\frac{4}{3} \pi R_I^3 n_H = S_* \Delta t$$

$$R_I = \left(\frac{3}{4\pi} S_* \Delta t \right)^{1/3}$$

*

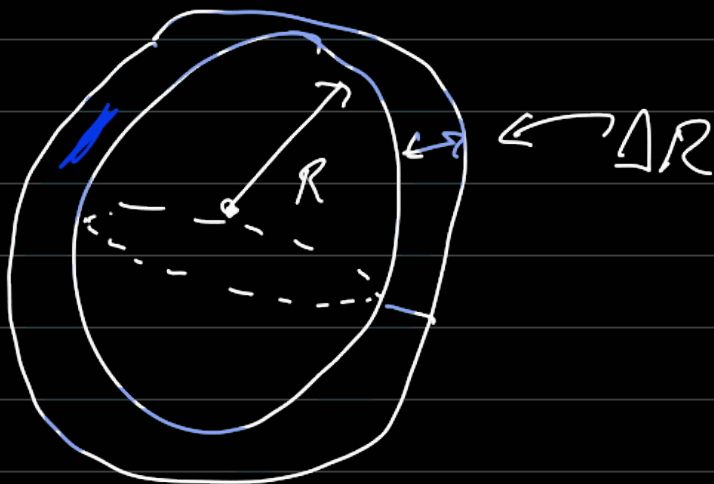
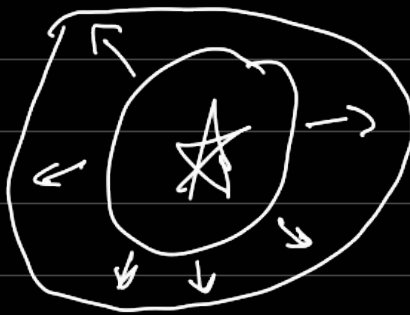
$R_I = R_I(t)$ is a function of time so the ionization region increases as time passes.

R_I : Defines HII region

Q6 star: $L = 20 L_\odot$, $S_* \approx 10^{49} \text{ s}^{-1}$
 $n_H \approx 10^9 \text{ m}^{-3}$

~> After $t = 10,000 \text{ yrs}$
 $R_I \approx 4 \times 10^{16} \text{ m} \approx 3 \text{ pc}$

* Start w/ an R_I



$$N_r = \checkmark n(r) = S_* \Delta t$$

$$V = 4\pi R^2 \cdot \Delta R$$

$$N_r = 4\pi R^2 \Delta R n(R) = S_* \Delta t$$

$$\frac{\Delta R}{\Delta t} = \frac{S_*}{4\pi R^2 n(R)}$$

$$\lim_{\Delta t \rightarrow 0} : \boxed{\frac{dR_I}{dt} = \frac{S_*}{4\pi R^2 n(R)}}$$

• Now, text assumes $\frac{n_H}{n_0} = \left(\frac{R}{R_0}\right)^{-3/2}$
 (can see this directly w/ "advanced" look at things)

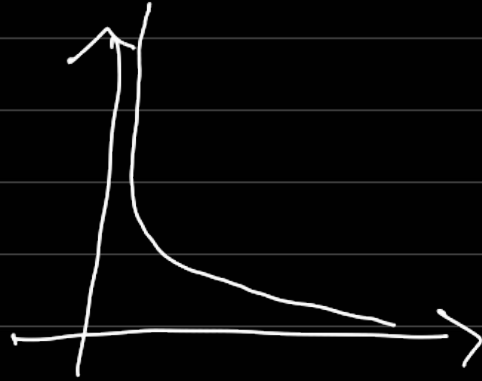
$$1/3 \quad S_* \propto 1/3$$

$$R_I(t) = \left(\frac{3}{8\pi} \frac{c^2}{n_0 R_0^{3/2}} \right)$$

$$R_I(t) \sim t^{2/3}$$

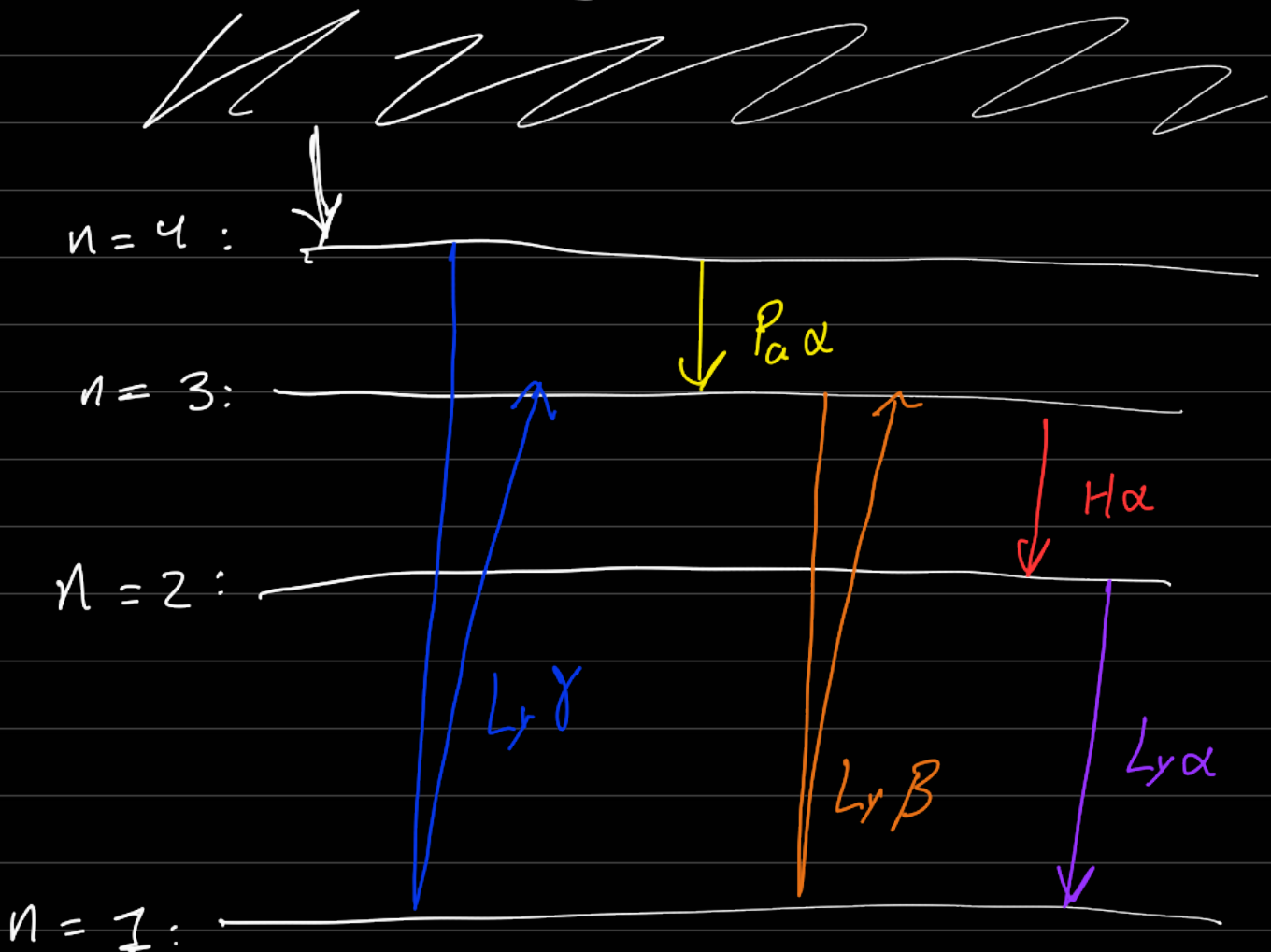
• non-uniform grows twice as fast
 $R_0^{3/2}$

$$n(R)_n = n_0 \frac{R_0^{3/2}}{R^{3/2}}$$



R_0 : usually, at extremes $(R_\star)$

Radiative Recombination:



H α : $\lambda \approx 6563 \text{ \AA}$ $3 \rightarrow 2$
(Most prominent in our sun $\odot$)

$$E_T = E_e + E_k$$

$\uparrow$
K.E.

* Ly α : $1215.67 \text{ \AA}$: hits dust
 121 nm

• Need a balance between ionization and recomb.

* $R \equiv$ rate of collisions/recomb.

$$R \propto n_p \quad \text{radiative recomb}$$
$$R = \alpha n_p \quad \alpha: \text{coeff.}$$

$$[R] = \frac{\# \text{ recomb}}{\text{sec}}$$

$$t_{\text{rec}} = \frac{1}{R} = \frac{1}{\alpha n_p}$$

R : for a single

$$R_{\text{ne}} \approx \frac{\# \text{ of recomb}}{\text{volume}} = \frac{R_{\text{tot}}}{V}$$

$$R_{\text{tot}} = V R_{\text{ne}} = \frac{4}{3} \pi R_{\text{F}}^3 \alpha n_p n_e$$

(rate of recomb)

For equilibrium:

$$S_* = R_{tot} = \frac{4}{3} \pi R_S^3 \propto n_p n_e$$

$$n_e \approx n_p$$

$$R_S \equiv \left(\frac{3}{4\pi} \frac{S_*}{\alpha n_H^2} \right)^{1/3}$$

$$* \quad n_H(R_f) 4\pi R_f^2 \Delta R$$

$R_{tot} \Delta t = \#$ of recomb.

$$R_{tot} = \frac{4\pi}{3} R_I^3 n_e \propto n_p$$

S_{*} was const.

$$4\pi R_I^2 n_H(R_H) \Delta R_I = S_* \Delta t - \frac{4}{3} \pi R_I^3 \alpha n_H^2 \Delta t$$

$$4\pi R_I^2 n_H(R_I) \frac{dR_I}{dt} = S_* - \frac{4}{3} \pi R_I^3 \alpha n_H^2(R_I)$$

$$\frac{dR_I}{dt} = 0 \quad (\text{no longer extends})$$

$$R_I = R_S = \left(\frac{3}{4\pi} \frac{S_*}{\alpha n_H^2} \right)^{1/3}$$

(Max size of HII region)

$$3R_I^2 \frac{dR_I}{dt} = (R_S^3 - R_I^3) \alpha n_H$$

$$\lambda \equiv R_I / R_S$$

$$\tau \equiv t / t_{\text{tran}}$$

$$3\lambda^2 \frac{d\lambda}{d\tau} = 1 - \lambda^3$$

$$\frac{d\lambda^3}{d\tau} = 1 - \lambda^3$$

$$\omega = \lambda^3 \quad \int \frac{d\omega}{1-\omega} = \int d\tau$$

$$\bullet - \ln(1-\omega) = \tau + C$$

$$1-\omega = \exp(-\tau)$$

$$\omega = 1 - e^{-\tau}$$

$$\lambda^3 = 1 - e^{-\tau} \Rightarrow R_I(t) = R_S^3 (1 - e^{-t/t_{rec}})$$

$$R_I(t) = R_S (1 - e^{-t/t_{rec}})^{1/3}$$

$$t \rightarrow \infty \quad R_I(t) \rightarrow R_S$$

$$R_I(0) = 0$$

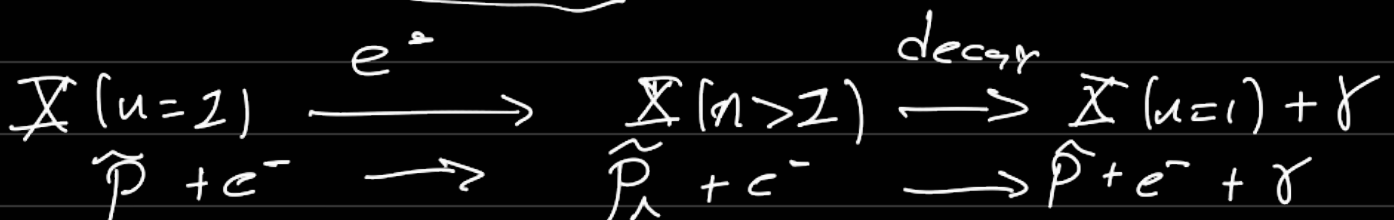
↳ Not as realistic, want a time t_0 s.t.

$$R(t_0) = R_*$$

$$\text{or, coll} \quad R_I(0) = 0 = "R_*$$

Temp in Nebulae:

Collisional Excitation:



• Convert kinetic into radiative energy

$$P(n) = \frac{1}{Z} e^{-E/kT}$$

↑ n^*

$$E_\gamma + E_{n_2} = E_{n_1} \quad \rightsquigarrow \quad \perp_{n=1}$$

$$E_r = E_{n_1} - E_1$$

↑ or K.E.

$$\bullet \quad p_e(n) = \frac{e^{-E_e/kT}}{Z} = \frac{e^{-\frac{(E_n - E_1)}{k_B T}}}{Z}$$

as claimed

• Now, $e^{-E_e/kT}$ gets smaller as E_e is larger, so we see less lines with $n' \rightarrow n=1$ for $n' \gg 1$

$$\bullet \quad \begin{array}{c} \text{----- } n'' \\ \text{----- } n' \end{array}$$

$$\text{----- } n=1$$

$$\frac{P(E'' - E_1)}{P(E' - E_1)} = e^{-(E'' - E')/k_B T}$$

$$I \propto P$$

$$dI \approx \underline{P(E)}(\bullet) dE$$

$$\frac{I_{n'}}{I''} = e^{\Delta E/kT} \quad \Delta E = E'' - E'$$

•

$$\text{Maxima: } \langle E \rangle = k_B T$$

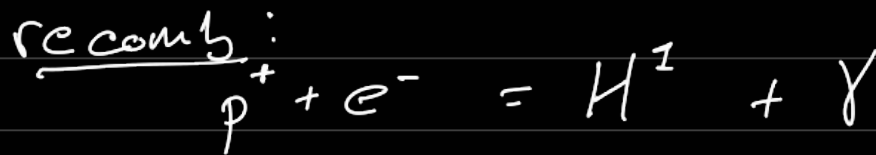
$$G = S_{\star} \cdot R_{T\star} : \left(\frac{\# \text{ photons}}{\text{Sec}} \right) \cdot \text{Energy of 1 ph}$$

$$G : \frac{\text{Energy}}{\text{s}} \approx \text{"Power" / Luminosity}$$

Cooling: Name for loss of energy.

• radiative recomb: K.E. transferred to radiative energy

$$[R_{\text{tot}}] \frac{\# \text{ of recomb}}{\text{s}}$$



$$\langle E_x \rangle = k T_e$$

$k T_e R_{\text{tot}}$: Energy radiated away

$$\underline{k T_e \left(\frac{4}{3} \pi R_I^3 \alpha n_H^2 \right) = L_G}$$

In eq.

$$T_e = T_{\star}$$

So must have separate forms of temp.
 * look at ϕ absorption/emission

* Cooling via Collisional Excitation:

i) Neutral H: emits $L\alpha$ $\lambda 1216$
 + $H\alpha$ $\lambda 6563$

ii) OII (O^+): emits ...

iii) $OIII$ (O^{++})

iv) NII (N^+)

* Main coolants for $H II$ regions is the emission from $O II$ (O^+), $O III$ (O^{++}), $N II$ (N^+)

